Second-Order Visual Attention Prediction in Multiagent Videos via Emotion-Aware Belief Inference

Chenglizhao Chen, Xinyu Liu, Mengke Song, Guotao Wang, Shanchen Pang, Hong Qin, Jufeng Yang, and Hui Yu

***Abstract*—In human social cognition, understanding others' attention relies not only on explicit cues such as gaze and posture but also on observers' inferences of implicit factors, including emotions, task goals, and social roles. This belief-based ability, known as second-order attention modeling, is a key component of theory of mind (ToM), enabling individuals to infer others' attentional states during multiagent interactions. Existing vision systems mainly focus on first-order attention modeling, such as gaze estimation and gaze following, which predict physical gaze locations. Although effective at the perceptual level, these methods cannot capture belief formation or account for modulatory factors such as emotions, limiting their applicability to higher-level social cognition. To address this limitation, we propose a belief-driven framework for second-order attention modeling, formulating observer inference under uncertainty as conditional probability estimation and integrating emotional cues, task semantics, and social roles. The framework extends attention modeling from gaze perception to emotion-modulated belief reasoning, enabling prediction of observer-reported attentional beliefs from visual, visually inferred affective, and social-context cues when direct gaze evidence is limited or ambiguous. We further introduce a large-scale dataset with diverse agents, tasks, and emotional contexts, whose observer-reported labels are validated through test–retest analysis, together with an emotion-conditioned evaluation metric for social–cognitive assessment. Experiments demonstrate that the proposed framework outperforms existing methods and produces predictions consistent with observer reports, showing effectiveness for emotion-aware second-order attention modeling.**

***Index Terms*—Second-order attention modeling, belief representation, belief reasoning, emotion-aware gaze prediction.**

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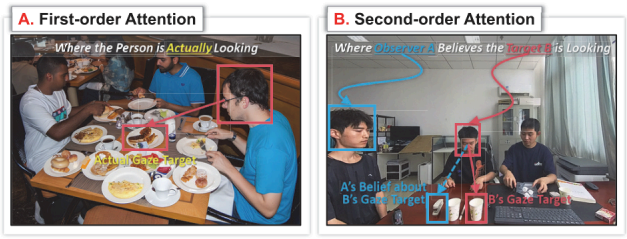


Fig. 1. First-order vs. second-order attention prediction. The left panel illustrates a first-order attention task, where the model predicts the object that person B is looking at on the basis of gaze cues. The right panel shows the proposed second-order attention task, which infers the object that observer A believes B is attending to. This belief may differ from reality because of A's perspective, emotional state, or social assumptions, emphasizing the subjectivity and uncertainty that are central to the theory of mind.

I. Introduction

UNDERSTANDING others' mental states is a core ability underlying social interactions and cooperation. Among these states, inferring where another person is looking is vital for interpreting their intentions and emotions. Humans do not rely solely on gazes or postures but instead build internal hypotheses about others' attentional foci, forming subjective beliefs shaped by experiences and emotions. Even when these beliefs deviate from reality, they can still influence behavior, e.g., when a person wrongly believes that someone is looking at them and responds to that gaze. Developmental psychology has shown that around four to five years of age, children develop second-order belief reasoning skills, allowing them to understand both others' true attention and their own beliefs about it. Classic paradigms such as the Sally–Anne experiment [1] and false-belief tasks [2] have identified this ability as a milestone in the development of the theory of mind [3], [4], enabling the representation of others' mental states in multiagent and partially observable situations. Such reasoning relies on a high-level belief construction procedure rather than direct perception and is modulated by emotions: positive affectations broaden attention, whereas anxiety or fear biases judgments.

In computer vision, most studies focus on first-order attention modeling, which involves predicting an individual's actual gaze target from cues such as their gaze direction [5], [6] or head pose [7], [8], as shown in Fig. 1-A. Although this research has advanced tasks such as head pose estimation, saliency prediction, and gaze following, it remains limited to estimating physical attention. However, an observer's behavior often depends on his or her belief about another person's

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TABLE I
COMPARISON BETWEEN FIRST-ORDER AND SECOND-ORDER ATTENTION MODELING

Dimension	First-Order Attention	Second-Order Attention
Objective	Predict actual gaze location (Where is B looking?)	Predict A's belief about B's gaze (Where does A believe B is looking?)
Representation Level	Perceptual (Physical facts)	Cognitive (Belief representation/Theory of Mind)
Cue Dependency	Explicit Visual Cues (Geometry, head pose, eye/face features)	Contextual & Visual Evidence (Emotion, goals, roles, visual evidence)
Inference Mechanism	Data-driven estimation (Based on explicit visual cues)	Belief-driven conditional inference (Based on understanding A's mental state)
Typical Tasks	Gaze estimation, saliency, joint attention (Physical focus)	Subjective attention reasoning ("I think you look at . . .")
Applications	Saliency analysis, HCI, behavior understanding	Collaboration, education, mental health, human-robot interaction
Strengths	Accurate physical gaze prediction	Subjective belief, closer to real social cognition

attention rather than their actual gaze position. To capture this psychological reasoning involving "what I think you are looking at", we introduce second-order attention modeling, as shown in Fig. 1-B. This task extends beyond the perceptual level and involves belief-based inferences, which are further influenced by emotional states, as emotions play a crucial role in shaping beliefs and attentional biases.

This distinction is not only conceptually important but also poses practical challenges in real-world scenarios. In industrial collaboration cases, operators often cannot directly observe their partners' gazes because of equipment-based occlusion and must infer attention from limited cues to coordinate their actions. If collaborative robots rely solely on actual gaze targets, their behaviors may contradict human belief-based judgments, causing communication delays, operational errors, or safety risks in assembly, medical, and emergency settings. Regarding mental health, individuals with anxiety or social disorders often show second-order attention biases, such as overestimating the likelihood that others are looking at them. These biases are shaped by emotional states, and without proper modeling, effective assessments and interventions become difficult to implement. In education and human–AI interactions, beliefs about others' attention influence engagements and interaction quality levels, while systems that ignore emotional and contextual cues fail to achieve psychological alignment. Overall, in multiagent and uncertain environments, behaviors are guided not by physical gazes but by belief-based attention that is shaped by emotions and context. The absence of such modeling poses efficiency and safety limitations and exposes a key gap in the social–cognitive reasoning capabilities of artificial intelligence. Therefore, developing socially intelligent vision systems requires advancing from low-level perception to a high-level reasoning scheme that enables machines to understand how humans understand others.

This study takes a key step toward bridging the gap between perceptual-level estimation and cognitive-level reasoning. We propose a belief-driven framework for conducting second-order attention modeling to advance the research on the visual theory of mind (ToM). In this framework, the second-order attention task is formally defined as predicting the region that observer A believes subject B is attending to within a scene. Unlike first-order gaze estimation, which involves predicting the target person's actual attended object, our task concerns predicting the observer-reported belief regarding what observer A thinks target B is attending to. This formulation is not intended as unconstrained mind reading from a single image but rather as a constrained visual–social prediction problem that is defined by an observer–target pair, video context, candidate objects, and confidence-based belief annotations. Accordingly, model performance is measured in terms of agreement with observer-reported reference beliefs rather than the direct recovery of an objectively observable internal mental state. From a modeling perspective, the key challenge lies in predicting observer-reported belief representations rather than physical gaze facts, which can be formulated as a belief-driven conditional inference problem. Unlike first-order modeling, which relies primarily on geometric and appearance cues, second-order modeling emphasizes subjectivity and context dependence and aligns more closely with the social cognition mechanisms of humans. Table I summarizes the systematic differences between first-order and second-order attention modeling.

With regard to validating the proposed belief-driven framework for second-order attention modeling, the existing data resources remain insufficient. Most gaze and joint attention datasets provide only ground-truth gaze annotations without systematically recording observers' beliefs, making them unsuitable for this new task. To bridge this gap, we construct a large-scale social interaction dataset, namely, the EASOAD, covering multiagent, multitask, and multiemotion scenarios. It offers observer-centered annotations that capture subjective judgments of others' attention along with confidence scores, providing the first systematic data foundation for implementing belief-level modeling. We further introduce an emotion-conditioned evaluation metric that jointly measures attention distributions and emotion predictions, enabling more comprehensive evaluations of belief consistency and emotional sensitivity and establishing an assessment standard that is more aligned with human reasoning.

Experiments show that our approach outperforms first-order baselines across diverse and complex scenarios. Beyond providing higher accuracy, it reproduces belief alignment, attentional bias, and context sensitivity patterns that are observed in participant reports. These results show that the proposed model can predict observer-reported beliefs across diverse social contexts. More importantly, this task provides a new direction for advancing artificial intelligence from low-level perception to high-level cognition. By modeling "what I think you are looking at", visual systems can better align their predictions with human-reported attention judgments in multiagent interactions. Our main contributions are as follows.

- We propose a belief-driven framework for second-order attention modeling that integrates emotional modulation into belief reasoning, bridging computer vision and theory of mind.

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- We construct EASOAD, the first large-scale dataset covering multiagent, multitask, and multiemotion scenarios, with observer-centered annotations including subjective beliefs, emotion labels, confidence scores, and test–retest reliability.
- We introduce an emotion-conditioned metric that jointly evaluates belief consistency and emotional sensitivity, providing a cognitively grounded benchmark for second-order attention modeling.
- Extensive experiments demonstrate that our approach surpasses first-order baselines in accuracy while achieving human-like consistency and interpretability, highlighting the significance of second-order attention modeling for social intelligence. The codes, datasets, and results are available.[1]

II. Related Works

A. First-Order Attention and Joint Attention Modeling

First-order attention modeling involves predicting an individual's actual gaze location within a visual scene. The early approaches were primarily based on visual saliency [9], [10], estimating attention by computing center–surround contrasts of low-level features such as color, intensity, and orientation. With the development of deep learning, CNN-based saliency models have achieved significantly better alignment with human eye-tracking data [11], [12].

The GazeFollow dataset [13] advanced gaze prediction by integrating head poses and contextual cues in large-scale scenes. The subsequent work introduced dual-pathway architectures to model head and scene features separately [14], [15] and later employed transformer-based models to capture spatiotemporal dependencies [8]. More recently, multiperson joint attention has become an active research area, with graph neural networks modeling interagent relations and collective attention patterns [16], [17], [18].

Although the recently developed methods have made progress, key limitations remain. Most approaches rely on geometric and appearance cues, restricting the modeling process to objective gaze estimation instead of subjective belief reasoning. Their robustness also decreases in multiagent or occluded environments where true gaze information is unavailable. The current methods also lack hierarchical reasoning capabilities to capture individual attention interpretation differences. Datasets such as EyeDiap [19], MPIIGaze [20], and Gaze360 [21] provide only ground-truth gaze annotations and do not support the belief-level or emotion-modulated modeling processes that are required for higher-level social cognition.

B. Theory of Mind and Visual Intention Understanding

Research in psychology has shown that the theory of mind (ToM) [3], [4] is a core cognitive ability that enables humans to understand others' beliefs and intentions under uncertainty. Classic experiments such as the Sally–Anne test [1] and cross-cultural studies have demonstrated that children approximately four to five years old begin to reason about others' false beliefs [22], indicating an important stage in social–cognitive development.

In the fields of computer vision and artificial intelligence, several studies have explored ToM-inspired reasoning methods. Some have introduced reinforcement learning or Bayesian frameworks to infer latent goals and agent behaviors [23], [24], whereas others have built visual commonsense reasoning datasets to infer intentions and contextual logic [25]. Recent visual intention understanding work has further advanced beyond low-level gaze or action recognition by combining visual cues with semantic priors. For example, CLIP-based intention understanding models infer human intentions through the synergy of visual perception and semantic reasoning [26], and performing holistic visual understanding with CLIP extends this direction to broader socially meaningful tasks such as intention recognition, emotion analysis, and scene understanding [27].

Although these studies have demonstrated the potential of intention-level visual reasoning, they do not address the core challenge of second-order attention. Most methods infer an agent's own intention, action goal, or semantic state rather than modeling what an observer believes another agent is attending to. In contrast, our work formulates an observer-centric second-order attention task and links low-level perceptual cues such as gazes, postures, and head appearances with high-level belief reasoning in a social context. This offers a new pathway for advancing the visual theory of mind.

C. Emotion-Conditioned Subjective Attention Modeling

Subjective attention modeling focuses on observers' judgments of others' attentional states rather than objective gaze locations [28], [29]. Attention bias theory in cognitive psychology suggests that emotional states systematically influence attention perception, underlying psychological phenomena such as social anxiety [30], [31].

In computer vision, this direction remains underexplored. Existing studies have investigated personalized saliency, individual differences in visual attention, emotional effects on gaze communication, and perspective-dependent attention prediction in multiperson interactions [32], [33], [34], [35]. However, these works are limited to controlled or small-scale settings and focus on perceptual attention rather than belief formation.

Recent multimodal studies have further demonstrated the importance of emotion-aware social understanding. Uncertain multimodal intention and emotion understanding models intentional and affective states under noisy conditions [36], while EmoLLM explores multimodal emotional understanding with large language models [37]. However, they focus on recognizing intentions or emotions, whereas our work investigates how observer A's emotion modulates the belief about target B's attended object. We therefore propose a unified framework integrating emotional modulation, task objectives, role dynamics, and confidence-based annotations for emotion-driven second-order attention modeling.

D. Datasets for Attention and Social Cognition

Datasets provide the empirical foundation for attention modeling and social cognition research. Early first-order datasets focused on controlled gaze estimation, including EyeDiap [19] for RGB-D gaze estimation and Columbia Gaze [38] for gaze classification. With deep learning development, MPIIGaze

[1] https://github.com/LiuXY3366/EA-SOA

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TABLE II
COMPARISON AMONG THE EXISTING DATASETS FOR ATTENTION AND SOCIAL COGNITION

Datasets	Subjects / Scale	Social Interaction	Emotion Labels	Observer Belief	Primary Task
Columbia Gaze[2013] [38]	56 subjects, 5 head poses	No	No	No	Eye Contact Detection
EyeDiap[2014] [19]	16 subjects, RGB-D	No	No	No	Controlled gaze estimation
GazeFollow[2015] [13]	122K images	Yes (static image)	No	No	Gaze following in images
GazeCapture[2016] [39]	2.5M images, 1450 subjects (mobile devices)	No	No	No	Mobile gaze estimation
MPIIGaze[2017] [20]	15 subjects, 213K images	No	No	No	Appearance-based gaze regression
VideoCoAtt[2018] [42]	492K video images	Yes (with audio, speaking activity)	No	No	Co-attention & interaction reasoning
Gaze360[2019] [21]	238 subjects, in-the-wild (indoor & outdoor, 360°)	Limited	No	No	360° gaze estimation
Social-IQ[2019] [43]	1,250 social videos	Yes	No	No	Social interaction Q&A, social reasoning
VideoAttentionTarget[2020] [40]	164K images	Yes (multi-person, temporal)	No	No	Video-based gaze following
AVA-ActiveSpeaker[2020] [41]	38.5 hours of videos	Yes (multimodal, speaking activity)	No	No	Active speaker detection (attention related)
EASOAD (Ours)	**1,378 videos, multi-task, multi-affect**	**Yes (multi-person, multi-task)**	**Yes**	**Yes (with confidence)**	**Second-order attention modeling**

[20], GazeCapture [39], and Gaze360 [21] further advanced appearance-based gaze estimation from laboratory settings to in-the-wild scenarios.

For social and multiperson interactions, GazeFollow [13] introduced gaze target annotations, while VideoAttentionTarget [40] extended the task to videos. Other datasets, such as AVA-ActiveSpeaker [41], VideoCoAtt [42], and Social-IQ [43], further explored multimodal social understanding and group interactions.

However, existing datasets mainly focus on objective gaze targets and do not capture observer beliefs or affective and role-related factors required for subjective attention modeling. To address these limitations, we construct EASOAD, a benchmark with diverse interactions and belief-centered annotations including confidence ratings. A comparison with existing datasets is provided in Table II.

III. BENCHMARK

A. Task Definition

In social cognition tasks, understanding others' attentional foci is essential for inferring their intentions, emotions, and beliefs. Attention modeling can be divided into first-order and second-order cases. We first introduce first-order attention modeling and then describe second-order modeling, highlighting its added complexity and distinctions.

First-Order Attention Modeling. The goal of first-order attention modeling is to predict the actual gaze target of a subject (e.g., person A), which is typically based on observable visual cues such as head poses and eye movements. Formally, first-order tasks can be defined as follows:

$$\hat{y}_A = f_\phi(I, \mathbf{P}_A), \tag{1}$$

where $\hat{y}_A$ represents the actual gaze target of person A, I is the scene image containing the subject's visual information, and $\mathbf{P}_A$ represents the position and pose of person A. f_ϕ is the model used to predict the gaze target from visual cues. First-order tasks involve predicting "objective facts", where the gaze target is observable and inferred from visual cues.

Second-Order Attention Modeling. In contrast with first-order attention modeling, second-order attention modeling is aimed at inferring observer A's belief about the attentional focus of person B. Psychological research indicates that second-order attention reflects a higher-level mental state about others' minds. According to Perner's theory of metarepresentation [44], its key feature is the ability to form beliefs about another individual's psychological state, such as "A believes that B is attending to a specific object", whereas first-order attention concerns only the direct perception of another's attention, such as "A knows that B is attending to that object".

Second-order attention modeling involves predicting what observer A believes person B is attending to from A's perspective. Under the observer-centric setting, A has no direct access to B's internal emotional state or ground-truth emotion label but can use visually observable cues derived from the scene, including B's head pose, facial appearance, and visible affective cues. This belief is shaped not only by visible cues but also by A's emotions, cognitive state, task goals, and social roles. Formally, second-order attention modeling can be defined as follows:

$$\hat{y}_{A\to B} = f_\theta(I, \mathbf{P}_A, \mathbf{P}_B, E_A, \tilde{E}_B^{vis}), \tag{2}$$

where $\hat{y}_{A\to B}$ represents the predicted attention target of person B as inferred by observer A; I is the scene image; $\mathbf{P}_A$ and $\mathbf{P}_B$ are the positions and poses of A and B, respectively; E_A encodes A's emotional or cognitive state; and $\tilde{E}_B^{vis}$ denotes the visible affective cue of target B rather than B's internal emotional state. Specifically, $\tilde{E}_B^{vis}$ is estimated from B's visible head/face region by a pretrained residual masking network (RMN)-based facial expression recognizer [45].

In this task, E_A denotes a conceptual observer-affective conditioning variable; in EASAPNet, it is instantiated by e_A inferred from A's visible head/face appearance rather than by a ground-truth emotion label at inference. It is encoded following Ekman's basic emotion theory [46] and represented as a multidimensional vector that models the probability distribution over seven emotions [47]: happiness, sadness, anger, fear, disgust, surprise, and neutrality. Emotions influence the reasoning process and affect attention inferences. For instance, positive emotions broaden A's attentional scope, whereas anger or fear may restrict it. Notably, E_A is the subjective affective state that modulates the belief inference process, whereas $\tilde{E}_B^{vis}$ is only appearance-based visual evidence that is

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inferred from B's observable region and does not imply access to B's internal mental state.

The key distinction between first-order and second-order attention modeling lies in their prediction targets. First-order modeling predicts an individual's actual gaze target from visual cues and treats it as an objective fact. In contrast, second-order modeling infers observer A's belief about the attentional focus of person B. This belief is shaped by A's emotions, social roles, and task goals, introducing inherent subjectivity. During this process, A's internal emotion serves as a subjective modulator of belief formation, whereas B's visible affective cues are used only as observer-accessible visual evidence.

While first-order tasks emphasize objective gaze prediction, second-order attention requires reasoning beyond perceptual cues by incorporating emotional, social, and contextual information. Although both tasks use B-centric perceptual evidence, they are not equivalent when A's belief dissociates from B's first-order attention reference or when belief changes under matched B-centric evidence with different A-specific conditions. We verify these properties in Section V-D11.

Accordingly, our evaluation compares predictions with A's observer-reported belief labels rather than B's actual gaze target. These labels represent A's reported belief and are not treated as objectively observable ground truth of a latent mental state. Therefore, prediction accuracy measures agreement with A's report rather than direct recovery of an internal state independent of that report.

Our formulation is inspired by theory-of-mind reasoning but should not be interpreted as a false-belief benchmark. Instead, it operationalizes second-order visual belief reasoning as predicting A's reported belief about B's attention under limited perceptual access, emotional states, and social context. False belief-like dissociation is considered a subset of the task rather than a requirement for every sample.

For second-order tasks, the evaluation process focuses not only on prediction accuracy (e.g., the MSE or IoU) but also on belief consistency, which measures whether the predicted results align with the observer's belief pattern. This distinction highlights a fundamental difference between the evaluation criteria of second-order and first-order attention modeling.

Task Scope and Applications. Second-order attention modeling is applied to scenarios that require subjective reasoning and social cognition capabilities, such as large-scale interactions or emotionally complex tasks. For instance, a teacher judging whether a student believes that their desk mate is attending to the blackboard or a phone reflects a belief inference. In contrast, tasks such as monitoring a camera to see where a pedestrian is looking fall under first-order modeling, which focuses on objective gaze prediction. Second-order modeling is less relevant when only visual cues matter, such as gaze prediction in scenes without observers, where emotional and social factors play little role.

B. Emotion-Aware Second-Order Attention Dataset

To support research on emotion-aware second-order attention reasoning, we built an emotion-aware second-order attention dataset (EASOAD). The dataset examines how observers infer others' attentional foci under different emotional states, advancing the study of emotion-modulated attention. Unlike the traditional gaze estimation or gaze-following tasks, the EASOAD moves beyond gaze direction prediction to model the role of emotion in reasoning scenarios. It ensures reliability and ethical compliance through anonymization, informed consent, and cross-validation to verify the consistency and reproducibility of labeling results. The collection protocol further includes validated emotion induction, observer-centric belief reporting, and independent quality-control verification schemes to ensure that the final labels correspond to observer A's own belief.

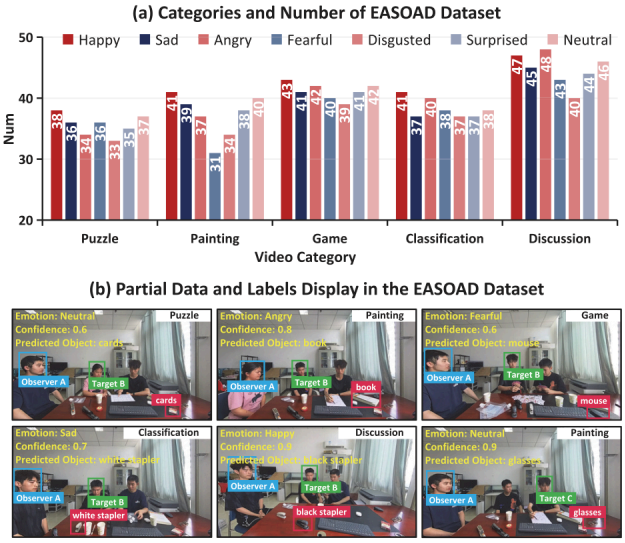


Fig. 2. (a) Category and emotion distributions of the EASOAD across five social tasks. (b) Partial data and labels contained in the EASOAD, illustrating observer emotions, confidence scores, predicted objects, and corresponding annotations.

1) Dataset Statistics: The EASOAD is a large-scale emotion-aware second-order attention dataset containing 1,378 videos across five social task categories, namely, Puzzle, Painting, Game, Classification, and Discussion, as shown in Fig. 2(a). Each category includes varied interaction scenarios to ensure the richness and complexity of the data. The Puzzle task includes three difficulty levels with 117, 82, and 50 samples, totaling 249 samples. The Painting task includes creation and evaluation subtasks with 260 samples. The Game task covers strategy, card, and board games with 288 samples. The Classification task contains four classification scenarios with 268 samples, and the Discussion task consists of dyadic free discussions with 313 samples. These tasks vary in their interaction complexity levels and contextual semantics, and each task includes seven affective conditions for the observer, i.e., happy, sad, angry, fearful, disgusted, surprised, and neutral, ensuring broad affective and social coverage.

In addition, the EASOAD fully considers the diversity and balance of its samples across multiple dimensions. Approximately 60% of the samples are fully visible, 30% are partially occluded (with head occlusion areas below 50%), and 10% are heavily occluded (with head occlusion areas greater than or equal to 50%). With respect to viewpoints, 50% of the samples were captured from the frontal view, 30% were obtained from the side view, and 20% were derived from the top-down view. The 40 participants, with one designated observer A providing the real-time belief annotation for each

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clip, ranged in age from 18 to 32 years, with an average age of 25. The gender distribution was 60% male and 40% female. In terms of their educational backgrounds, 45% of the participants were undergraduate students, 35% were master's students, and 20% were doctoral students. These statistical characteristics ensure that the EASOAD maintains strong representativeness, structural balance, and contextual diversity, providing a reliable foundation for conducting emotion-aware second-order attention modeling.

2) Dataset Collection: The data collection process for the EASOAD followed a standardized experimental procedure. The participants were recruited through social platforms, signed informed consent forms, and provided basic personal information. During emotion induction, the participants watched validated emotion-eliciting film clips selected from FilmStim [48]. The clips were selected according to their normative affective ratings and mapped to the seven affective conditions used in the EASOAD: happy, sad, angry, fearful, disgusted, surprised, and neutral.

Each induction trial consisted of a neutral baseline period, an affective film clip, a manipulation check, and the subsequent second-order attention annotation task. Immediately after each clip, the participants completed self-assessment manikin (SAM) ratings and a discrete emotion check to verify the effectiveness of the induction process. A trial was retained only when the reported dominant emotion matched the intended category and the target emotion intensity level exceeded 3; otherwise, the trial was not used in the final dataset.

We ensured that all the participants followed the same emotion induction process to eliminate potential biases. Only trials that passed the emotion induction check were used for observer-centric belief annotation purposes. After emotion induction and verification, the same participant acted as observer A and was recorded while performing the subsequent social interaction task, rather than serving as an external annotator. Thus, the observer's visible appearance, induced emotion label, and real-time belief report corresponded to the same participant and trial. The observers inferred the attentional focus of the target individual (B or C) from their perspective. Each observer was informed of the current target (B or C) before the task began to stay focused and avoid confusion due to task switching. The interval between a successful induction and the annotation task was kept short so that the reported belief corresponded to the induced affective condition.

3) Dataset Annotation: During the annotation process, the EASOAD employed a structured annotation flow to ensure its consistency and accuracy. The second-order attention label of each sample was obtained from the real-time subjective report of the participant who was assigned to the role of observer A. This report recorded observer A's own belief under the current emotional state and social context rather than a third-party inference made after viewing the video. Each sample was annotated by observer A, who inferred the attentional focus of the target individual (B or C) on the basis of their own perspective, emotional state, and social context. During each annotation round, the target individual was specified in advance, and the system prompted observer A to report their attention inferences at fixed intervals (e.g., every 5-10 seconds). This approach guaranteed the standardization and consistency of the annotation process.

Specifically, at each designated time point, observer A selected the object on which they believed the target individual was focusing on using an electronic device. Candidate objects were generated by a preprocessing model (e.g., Grounding DINO [49]) with category labels and visual attributes. Observer A also assigned a confidence score (ranging from 0 to 1) to each judgment, reflecting their certainty in the inferred attention, as shown in Fig. 2(b). This report recorded observer A's own belief under the current emotional state and social context rather than a third-party inference made after viewing the video. Therefore, the recorded response of observer A served as the observer-reported reference belief label for second-order attention modeling rather than an objectively verified ground truth of A's latent mental state. The emotional state of observer A and the visible affective cue of the target individual (B or C) were concurrently annotated to examine how emotions influence attention inferences. Here, the target-side annotation described apparent affective cues that were observable from facial appearances and behaviors rather than the target individual's inaccessible internal emotional state.

All the information, including observer A's emotional label, the target's visible affective label, the attended object, and the confidence score, was mapped to the corresponding image frame, forming structured data for the subsequent model training and evaluation stages. The target-side visible affective label was used only to describe observable affective evidence or as auxiliary supervision for the visual affect encoder; during the inference procedure, it was not provided to the second-order attention prediction model as the target individual's true internal emotion. This structured annotation process ensured annotation consistency and reliability, providing high-quality labeled data for an emotion-driven attention reasoning analysis.

To verify the consistency of the annotations, all the recorded belief reports were checked by three independent quality-control annotators. These annotators did not provide second-order attention labels. Instead, they checked the target identities, frame alignment statuses, candidate–object mappings, annotation completeness, and consistency between the recorded responses and the final object label. The kappa coefficient [50] was computed to assess the agreement between the quality-control decisions and verified object–label mappings among the three annotators. The average kappa score was 0.85, indicating strong quality-control consistency. This value evaluated the agreement between quality-control decisions and object–label mappings rather than the temporal stability of observer A's subjective belief; the latter is evaluated separately in Section V-D10. For samples with inconsistent quality-control decisions or ambiguous object mappings, two cognitive psychology professors reviewed the records and decided whether to correct the mappings or exclude the samples. They did not replace observer A's subjective belief report. We also applied confidence filtering, retaining samples with scores of 0.6 or higher to ensure their reliability. This removed 12.3% of the original reports while retaining medium-confidence (0.6–0.7) cases. The final reference label consisted of observer A's belief report, the corresponding confidence score, and the verified object mapping. To reduce fatigue, 5-minute breaks

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were provided after every 30 minutes of annotation to maintain high quality.

4) Belief–Reality Dissociation Examples: We further identified representative belief–reality dissociation cases. In the perceptual-access dissociation case, B's gaze-based target was partially occluded from A's viewpoint, and A reported a different visible object as B's attended target. In the gaze-intention dissociation case, B's visible gaze or head direction pointed to one object, whereas the task-relevant intended target was another object indicated by the interaction context. In the emotion-modulated dissociation case, similar visual evidence led to different observer–belief distributions or confidence scores under different emotional states. These cases show that the EASOAD contains samples beyond direct subjective gaze–target estimation samples.

5) Ethical Compliance: The EASOAD collection process strictly followed ethical guidelines. All participants signed informed consent forms stating that their data would be used only for academic research. During the data collection and processing phases, all personal information, including facial features and private items such as phone screens, was anonymized to protect the privacy of the participants. The participants also retained the right to withdraw their data at any time. All facial images, interaction videos, and personal information were encrypted and deidentified to ensure their confidentiality.

C. Evaluation Metric

1) Background and Motivation: In second-order attention modeling, the task involves predicting both observer A's belief about subject B's gaze target and observer A's emotional state. These two aspects are strongly interrelated. Psychological studies have shown that emotions systematically modulate subjective inferences; for instance, positive affectations broaden attention, whereas anxiety biases judgments. Therefore, conventional metrics such as cross-entropy or KL divergence are inadequate, as they assess distribution similarity but ignore how emotion prediction errors influence belief inferences. Without a unified metric, a model may obtain an inflated score even when its gaze belief is accurate but its emotion alignment result is not accurate. This calls for an evaluation measure that jointly reflects belief consistency and emotion-driven modulation.

2) Metric Definition: We propose the emotion-conditioned optimal transport (ECOT) metric, which integrates belief and emotion predictions into a unified framework to assess their agreement with observer-reported beliefs under affective modulation.

Let $\mathcal{Y}$ denote the set of candidate gaze targets. The model outputs are as follows. $\hat{p}(y)$ is the predicted distribution that "A believes that B attends to y"; $\hat{E} \in \Delta^{K-1}$ is the predicted distribution of A's emotional state, with K emotion categories. The reference annotations include $p(y \mid E)$, observer A's reported belief distribution under the validated affective condition E, and $E \in \Delta^{K-1}$, which is A's validated reference emotion distribution.

Before computing ECOT, the predicted heatmap $\hat{H}$ and the observer-reported reference belief heatmap H^{gt} are normalized into probability distributions $\hat{p}$ and p, respectively. Specifically, we add a small smoothing constant ϵ_0 to avoid the numerical instability caused by zero probabilities:

$$\hat{p}(y) = \frac{\hat{H}(y) + \epsilon_0}{\sum_{y' \in \mathcal{Y}} (\hat{H}(y') + \epsilon_0)},$$
$$p(y|E) = \frac{H^{gt}(y) + \epsilon_0}{\sum_{y' \in \mathcal{Y}} (H^{gt}(y') + \epsilon_0)}. \tag{3}$$

The target discrepancy $d_Y(y, \tilde{y})$ is defined as the normalized Euclidean distance between the centers of the candidate targets or heatmap cells:

$$d_Y(y, \tilde{y}) = \frac{\|c_y - c_{\tilde{y}}\|_2}{\sqrt{W^2 + H^2}}, \tag{4}$$

where c_y and $c_{\tilde{y}}$ denote the centers of target y and target $\tilde{y}$, respectively, and W and H denote the width and height of the image, respectively. The emotion discrepancy $d_E(\hat{E}, E)$ is defined as the normalized L_1 distance between the predicted and reference emotion distributions:

$$d_E(\hat{E}, E) = \frac{1}{2}\|\hat{E} - E\|_1. \tag{5}$$

Since both $\hat{E}$ and E are probability distributions determined over emotion categories, this normalization process constrains $d_E(\hat{E}, E)$ to $[0, 1]$.

The emotion-conditioned cost function is defined as follows:

$$c_{\hat{E},E}(y, \tilde{y}) = d_Y(y, \tilde{y})^{\beta}\big(1 + \lambda\, d_E(\hat{E}, E)\big), \tag{6}$$

where $d_Y(y, \tilde{y})$ denotes the target discrepancy, $d_E(\hat{E}, E) \in [0, 1]$ denotes the emotion prediction error, $\beta > 0$ controls the degree of sensitivity to target mismatches, and $\lambda \geq 0$ regulates the amplification of emotional errors. In all the experiments, we set $\beta = 1$ and $\lambda = 0.5$ by default.

The minimal matching cost between the predicted and ground-truth distributions is then as follows:

$$W_c(\hat{p}, p(\cdot|E); \hat{E}, E) = \min_{\pi \in \Pi(\hat{p}, p(\cdot|E))} \sum_{y,\tilde{y}} \pi(y, \tilde{y})\, c_{\hat{E},E}(y, \tilde{y}), \tag{7}$$

where $\Pi(\hat{p}, p)$ is the set of couplings with marginals $\hat{p}$ and $p(\cdot|E)$. This cost quantifies the minimal effort that is required to align the predicted beliefs with the human beliefs under the predicted emotion.

During the implementation procedure, we use an entropic Sinkhorn solver to approximate the optimal transport problem for efficiently performing batch computations. The regularized transport cost is defined as follows:

$$W_c^{\varepsilon} = \min_{\pi \in \Pi(\hat{p}, p(\cdot|E))} \sum_{y,\tilde{y}} \pi(y, \tilde{y}) c_{\hat{E},E}(y, \tilde{y}) + \varepsilon \sum_{y,\tilde{y}} \pi(y, \tilde{y})(\log \pi(y, \tilde{y}) - 1). \tag{8}$$

We set the Sinkhorn regularization coefficient to $\varepsilon = 0.05$ and use 50 iterations.

Finally, ECOT is defined as follows:

$$\text{ECOT} = \frac{1}{N} \sum_{i=1}^{N} \exp\big(-W_c^{\varepsilon}(\hat{p}_i,\ p_i(\cdot|E_i); \hat{E}_i, E_i)\big), \tag{9}$$

with values in $(0, 1]$. Higher scores indicate stronger alignment between the predictions and emotion-shaped human beliefs. All testing samples are evaluated using the same distance definitions, hyperparameters, and Sinkhorn settings to ensure their reproducibility.

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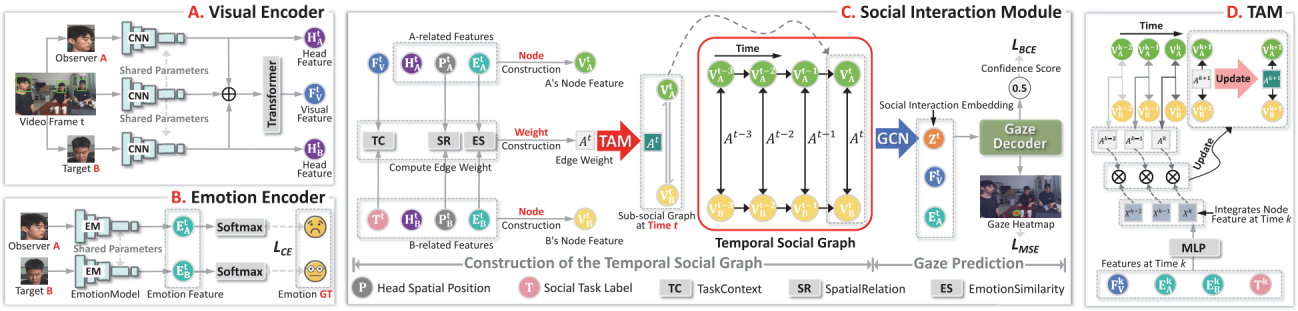


Fig. 3. Overview of EASAPNet. It is composed of four key modules: (A) a visual encoder for extracting scene and head features from video frames; (B) an emotion encoder for modeling the emotional states of the observer and the visible affective cues of the target; (C) a social interaction module, which builds a temporal social graph to capture dynamic social relations; and (D) a temporal attention mechanism (TAM) for adaptively aggregating temporal interaction information. The integration of these modules enables EASAPNet to jointly reason about visual cues, emotions, and social contexts for accurately predicting second-order attention.

3) Advantages: Compared with the traditional metrics, ECOT offers three advantages. It provides a unified evaluation system for belief and emotion prediction tasks without heuristic combinations. Its emotion sensitivity allows emotion errors to modulate belief evaluations in line with psychological findings. It also separates target and emotion discrepancies, offering clearer and more interpretable diagnostics of model behaviors.

Thus, ECOT serves as a compact, rigorous, and cognitively plausible benchmark for evaluating second-order attention results, offering a faithful reflection of social intelligence performance.

IV. Methods

A. Overview

On the basis of the EASOAD, we propose the emotion-aware social attention prediction network (EASAPNet), which predicts the gaze target of an individual in a social scene from the perspective of an emotionally influenced observer. Unlike the traditional gaze-following models, EASAPNet integrates visual cues, emotional states, and social interaction dynamics. Given a video clip, the emotional and positional information of observer A, and target B's visible affective cues, the model outputs a predicted gaze heatmap and a confidence score.

EASAPNet consists of four modules: (1) a visual encoder, (2) an emotion encoder, (3) a social interaction module, and (4) a temporal attention mechanism (TAM). The architecture of the network is shown in Fig. 3, and further details are provided below.

B. Visual Encoder

The visual encoder aims to extract joint representations of observer A and target B from video clips (Fig. 3-A). Given a video sequence $V = \{I^1, I^2, \ldots, I^T\}$, we first apply object detection and tracking methods to obtain the head coordinates of A and B in each frame:

$$\mathrm{Pos}_i^t = (x_i^t, y_i^t, w_i^t, h_i^t), \quad i \in \{A, B\}. \tag{10}$$

We subsequently crop the corresponding head images from each frame I^t:

$$\mathrm{Head}_i^t = \mathrm{Crop}(I^t, \mathrm{Pos}_i^t), \quad i \in \{A, B\}. \tag{11}$$

The image I^t, along with the cropped head images of A and B, are passed through a shared CNN to extract features:

$$f_I^t = \mathrm{CNN}(I^t); \quad f_i^t = \mathrm{CNN}(\mathrm{Head}_i^t), \quad i \in \{A, B\}. \tag{12}$$

These three features are concatenated and fed into a transformer to model the interactions between the scene semantics and human agents, yielding a unified visual embedding:

$$F_{\mathrm{visual}}^t = \mathrm{Transformer}([f_I^t; f_A^t; f_B^t]). \tag{13}$$

The resulting feature F_{visual}^t integrates contextual semantics, appearances, and relative spatial information, forming the foundation for the subsequent social attention reasoning process.

C. Emotion Encoder

Emotional states play a crucial role in subjective inference tasks (Fig. 3-B). The emotion encoder distinguishes observer A's subjective emotional state from target B's visible affective cue. We use a pretrained emotion recognition model, i.e., the residual masking network (RMN) [45] (g_ψ^{RMN}), to derive affect-related vectors from the head images in each video frame.

$$e_A^t = g_\psi^{\mathrm{RMN}}(\mathrm{Head}_A^t), \quad \tilde{e}_B^{vis,t} = g_\psi^{\mathrm{RMN}}(\mathrm{Head}_B^t), \tag{14}$$

where e_A^t denotes the emotional embedding of observer A and $\tilde{e}_B^{vis,t}$ denotes the visible affective cue estimated from the visible head/face region of target B rather than B's internal emotional state. Both embeddings are typically represented as probability distributions over seven basic emotions, including happiness, sadness, anger, fear, disgust, surprise, and neutrality. To enhance the representation learning procedure, we supervise the training process with observer emotion labels y_A^{emo} and target-side visible affect labels y_B^{app} derived from the dataset using the cross-entropy loss:

$$\mathcal{L}_{emo} = \mathrm{CE}(e_A^t, y_A^{emo}) + \eta \cdot \mathrm{CE}(\tilde{e}_B^{vis,t}, y_B^{app}), \tag{15}$$

where y_B^{app} is used only to supervise the learning of target-side visible affective cues and η controls the auxiliary supervision strength. The emotion labels in Eq. 15 are used only for training supervision. During standard inference, no emotion label is supplied; e_A and $\tilde{e}_B^{vis}$ are inferred from the visible head/face regions of A and B, respectively.

Finally, observer A's emotion feature, target B's visible affective cue, and visual embeddings are combined and fed

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into the social interaction modeling module, allowing the model to infer "where A thinks B is looking" under a specific observer's emotional state and target-side visible affective evidence.

The visual and emotion encoders combine perceptual cues and observer-centric affective cues to support social reasoning and attention prediction processes. This design more accurately models human cognition and advances visual understanding toward social intelligence.

D. Social Interaction Module

The social interaction module is a key component of EAS-APNet (Fig. 3-C). Integrating observer emotion, target-side visible affective cues, and task and behavioral cues, it refines observer A's gaze inferences in person scenarios. Unlike models that use only visual inputs, it leverages high-level semantics to produce more accurate and cognitively plausible predictions of A's inference regarding B's gaze.

1) Construction of the Temporal Social Graph: This module constructs a temporal social graph to represent the interactions between observer A and target B (or C). At each time t, the input consists of high-level spatial, pose, observer emotional, and target visible-affective features that are extracted by the vision and emotion encoders.

To comprehensively capture the dynamic interaction relationships in social scenes, this module constructs a temporal social graph G^t at each time step t, which is formally defined as follows:

$$G^t = \langle V^t, A^t \rangle, \tag{16}$$

where V^t represents the feature set of all nodes at time step t (i.e., observer A and target B), including spatial positions, poses, the observer emotion, target-side visible affective cues, and task information. A^t denotes the adjacency matrix at this time step, where each entry encodes the strength of the social relationship between nodes.

2) Node Construction: Each node feature vector V^t contains key information such as a spatial position, a head pose, and affect-related cues. For observer A, V_A^t consists of the spatial position P_A^t, head pose Head_A^t, and observer emotion e_A^t (extracted by the emotion encoder). For target individual B, the feature vector includes P_B^t, Head_B^t, $\tilde{e}_B^{vis,t}$, and an additional social task label T indicating the current behavioral type, where $\tilde{e}_B^{vis,t}$ denotes the visible affective cue estimated from B's visible head/face region rather than B's internal emotional state. Before conducting node fusion, heterogeneous modality features are first projected into a shared d-dimensional latent space. Specifically, the spatial position, head pose, observer emotion, target-side visible affective cues, and task context are mapped by modality-specific projection layers followed by LayerNorm and ReLU functions.

The node features are fused using a dynamic weighted strategy at each time step as follows:

$$V_A^t = \lambda_1 P_A^t + \lambda_2 e_A^t + \lambda_3 \text{Head}_A^t, \tag{17}$$

$$V_B^t = \lambda_4 P_B^t + \lambda_5 \tilde{e}_B^{vis,t} + \lambda_6 \text{Head}_B^t + \lambda_7\ T, \tag{18}$$

where λ_1 to λ_7 are learnable modality gates and are softmax-normalized within each node. This design enhances the contextual awareness and subjective gaze inference ability of the model in complex social scenarios while avoiding treating target-side affective evidence as the target individual's internal emotion.

3) Edge Construction: In the temporal social graph, edge weights represent the strengths of the social relationships between nodes, that is, the degree of interaction between observer A and target B (and optionally candidate C). The edge weights are computed on the basis of three main factors.

(1) Spatial Relation (SR): This quantifies the spatial distance and relative position between nodes. The location and pose of observer A affect gaze inference toward B, as closer or more salient individuals attract more attention. Thus, the spatial relation weight reflects the relevance of the spatial relation between A and B to A's inference of B's attention, which is typically computed by the Euclidean distance measure as follows:

$$\text{SR}(P_A^t, P_B^t) = \frac{1}{1 + \|P_A^t - P_B^t\|}. \tag{19}$$

(2) Apparent Affective Compatibility (AAC): This measures the compatibility between observer A's emotional state and target B's visible affective cue. Greater affective compatibility helps model how observer A interprets B's visible expression or behavioral cue under A's own emotional state when inferring B's gaze behavior. This term does not imply access to B's internal emotional state. Apparent affective compatibility is computed as the cosine similarity between the two affect-related vectors:

$$\text{AAC}(e_A^t, \tilde{e}_B^{vis,t}) = \frac{e_A^t \cdot \tilde{e}_B^{vis,t}}{\|e_A^t\| \|\tilde{e}_B^{vis,t}\|}. \tag{20}$$

(3) Social Task Context (TC): This models the influence of target B's current task on the gaze inference of observer A. To determine whether target B is engaged in a specific task, we compute the similarity between the unified visual feature F_{visual}^t and all candidate task embeddings T_k^t and normalize the similarity scores using the softmax function to obtain a probability distribution. This factor is defined as follows:

$$\text{TC}(T_j) = \begin{cases} 1, & \text{if the target task is most probable,} \\ 0, & \text{otherwise.} \end{cases} \tag{21}$$

Combining these factors, the adjacency matrix for the edge weights A_{ij}^t is constructed as follows:

$$A_{ij}^t = \alpha_1 \cdot \text{SR}(P_i^t, P_j^t) + \alpha_2 \cdot \text{AAC}(e_A^t, \tilde{e}_j^{vis,t}) + \alpha_3 \cdot \text{TC}(T_j), \tag{22}$$

where A_{ij}^t denotes the edge weight between node i and node j at time step t; i denotes the observer node; j denotes the target node; and P_i^t, $\tilde{e}_j^{vis,t}$, and T_j represent the spatial position, target-side visible affective cue, and task embedding, respectively. α_1, α_2, and α_3 are the weighting coefficients.

To further capture the temporal dynamics of social relationships, we introduce a temporal attention mechanism that adaptively adjusts the edge weights and dynamically integrates historical interaction information, enabling the model to represent the temporal evolution trends of social relationships. Compared with the traditional static weighting methods, this approach makes fuller use of observer emotions, target-side visible affective cues, visual features, and task contexts, achieving a more context-aware relationship modeling process in dynamic social environments.

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4) Temporal Attention Mechanism: The temporal attention mechanism (TAM, as shown in Fig. 3-D) updates the edge weights by aggregating historical interaction information. To avoid notational ambiguity, we distinguish the raw edge weight A_{ij}^k at historical time step k from the temporally updated edge weight $\tilde{A}_{ij}^t$ at the current time step t. The update is formulated as follows:

$$\tilde{A}_{ij}^t = \sum_{k=1}^{t} \beta_{t,k} \cdot A_{ij}^k \cdot \phi(\mathbf{X}_k), \tag{23}$$

where $\tilde{A}_{ij}^t$ denotes the temporally aggregated edge weight at the current time step t, A_{ij}^k denotes the raw edge weight between nodes i and j at historical time step k, $\beta_{t,k}$ is the temporal attention weight from time step k to time step t, and $\phi(\mathbf{X}_k)$ is a context-dependent gate that is computed from the features at the same historical time step k. The computations of $\beta_{t,k}$ and $\phi(\mathbf{X}_k)$ are given below.

Attention Weight $\beta_{t,k}$. The temporal attention weight $\beta_{t,k}$ is determined by the temporal distance between the current time step t and the historical time step k:

$$\beta_{t,k} = \frac{\exp[-\gamma(t-k)]}{\sum_{r=1}^{t} \exp[-\gamma(t-r)]}, \tag{24}$$

where γ is a hyperparameter for controlling the temporal decay process and $t-k$ is the interval between the current and historical steps. Historical interactions that are closer to the current time step receive higher weights. We use r as the summation index in the denominator to avoid ambiguity with the historical index k.

Adaptive Function $\phi(\mathbf{X}_k)$. The adaptive function integrates visual, observer emotional, target visible-affective, and task features at historical time step k. Specifically, the historical context vector is defined as follows:

$$\mathbf{X}_k = [F_{\text{visual}}^k, e_A^k, \tilde{e}_B^{vis,k}, T^k], \tag{25}$$

and the adaptive gate is computed as shown below:

$$\phi(\mathbf{X}_k) = \sigma(\text{MLP}(\mathbf{X}_k)), \tag{26}$$

where F_{visual}^k denotes the visual feature at time step k, e_A^k denotes the emotional state of observer A at time step k, $\tilde{e}_B^{vis,k}$ denotes the visible affective cue of target B at time step k, and T^k denotes the task context at time step k. Thus, all inputs of $\phi(\mathbf{X}_k)$ are taken from the same historical time step k, which removes the temporal inconsistency involved in the previous notation.

By leveraging this mechanism, the model dynamically adjusts the contributions of the historical edge weights through both temporal decay and context-dependent gating, thereby improving the temporal social relation modeling process. Compared with the traditional static weighting methods, this approach more precisely captures temporal dependencies and complex interactions, enhancing its adaptability to dynamic social environments.

5) Gaze Prediction: The gaze decoder in EASAPNet outputs a gaze heatmap P_{gaze} and a confidence score $\hat{c}$ for observer A toward target B, taking the input visual features F_{visual}, observer A's emotional feature E_A, and the aggregated graph features Z_t and dynamically updating them via graph convolution and self-attention to capture social interaction dynamics. These features are concatenated and passed through a fully connected layer with softmax activation to produce the gaze heatmap:

$$P_{\text{gaze}} = \text{softmax}\left(\text{FC}(F_{\text{visual}}, E_A, Z_t)\right), \tag{27}$$

where P_{gaze} denotes the predicted distribution of the scene regions that observer A believes target B is attending to. This is because only the internal emotion of observer A directly modulates their belief inference as a subjective affective factor; the affect-related representation of target B has already been incorporated into the graph feature Z_t as a visible affective cue rather than as B's internal emotional state.

The decoder also predicts a confidence score $\hat{c}$ for quantifying the certainty of the gaze prediction by passing the graph features Z_t through an MLP with a sigmoid activation function:

$$\hat{c} = \sigma\left(\text{MLP}(Z_t)\right). \tag{28}$$

6) Loss Function: During training, the loss function for the gaze prediction decoder consists of two main parts: a gaze heatmap regression loss (MSE) and a confidence regression loss (BCE). The overall loss is defined as follows:

$$L = \lambda_1 \cdot \text{MSE}(P_{\text{gaze}}, P_{\text{gaze}}^{gt}) + \lambda_2 \cdot \text{BCE}(\hat{c}, c^{gt}), \tag{29}$$

where λ_1 and λ_2 are weighting coefficients that adjust the importance of each loss component; P_{gaze} denotes the predicted gaze heatmap; P_{gaze}^{gt} is the observer-reported reference belief heatmap; $\hat{c}$ represents the predicted confidence score; and c^{gt} is the observer-reported reference confidence score.

V. Experiments

A. Dataset and Evaluation Metrics

1) Dataset: All the experiments are conducted on the proposed EASOAD, the first large-scale benchmark specifically designed for second-order attention modeling. It encompasses five realistic social tasks and features observer-centered subjective belief annotations with systematically recorded emotional states and confidence scores, offering essential data for investigating how emotions modulate subjective attention. The dataset is divided into 70%, 15%, and 15% splits for training, validation, and testing purposes, respectively. The split is participant-disjoint (28/6/6 participants), with all clips from each participant assigned to a single subset, although physical scenes are not strictly disjoint across subsets.

2) Evaluation Metrics: To comprehensively evaluate the performance of the developed model, we adopt dimensionwise metrics. First, three standard metrics are used to measure the consistency between the predicted and observer-reported reference belief heatmaps: the mean squared error (MSE), Kullback–Leibler (KL) divergence, and intersection-over-union (IoU). Second, to assess the social–cognitive reasoning results, we employ our proposed emotion-conditioned optimal transport (ECOT) metric, which jointly evaluates belief–emotion consistency through a cost function integrating emotion errors into the distance computation, thereby quantifying the degree of agreement with emotion-conditioned observer reports. A higher score indicates stronger agreement with the reference belief annotations.

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TABLE III

QUANTITATIVE COMPARISON BETWEEN OUR PROPOSED METHOD AND THE SOTA GAZE-FOLLOWING METHODS ON THE EASOAD

Methods	MSE↓	KL↓	IoU↑
MSGF	0.127	0.263	0.411
VideoAtt	0.123	0.249	0.418
DualAtt	0.119	0.232	0.429
HGTTR	0.122	0.215	0.427
ViTGaze	0.112	0.194	0.452
Gaze-LLE	0.105	0.177	0.468
EASAPNet (Ours)	**0.084**	**0.120**	**0.523**

3) Implementation Details: Our model is implemented in PyTorch and trained on a NVIDIA GeForce RTX 4090 GPU. The visual encoder adopts a pretrained ResNet-50 model [51], and the emotion encoder is initialized with a pretrained RMN model and fine-tuned. We use AdamW with an learning rate of 1×10^{-4} and cosine annealing. The batch size is 32, and the model is trained for 100 epochs. The loss weights ω_1 and ω_2 are set to 1.0 and 0.5, while the social graph weights α_1, α_2, and α_3 are set to 1.0. During inference, the model does not use B's ground-truth emotion label; instead, $\tilde{e}_B^{vis}$ is predicted from B's visible head/face crop and serves as observer-accessible evidence rather than privileged information. All results follow this observer-centric protocol.

B. Comparison With the SOTA Gaze-Following Methods

To validate the effectiveness of EASAPNet in the second-order attention task, we compare it with six state-of-the-art (SOTA) gaze-following methods (MSGF [52], VideoAtt [40], DualAtt [53], HGTTR [54], ViTGaze [55], and Gaze-LLE [56]). As these baselines were designed for first-order tasks, we retrain them on the EASOAD using only traditional inputs (e.g., head poses and scene images) and evaluate their predictions against our second-order belief annotations. Therefore, Table III should be interpreted as a system-level comparison, since the observed margins may reflect both additional conditioning information and architectural differences.

1) Quantitative Analysis: As shown in Table III, EASAPNet outperforms all the other methods in terms of all the metrics (the MSE, KL divergence, and IoU), demonstrating a clear advantage. Since the first-order baselines do not incorporate observer emotion or task/social context, this difference should not be attributed solely to architecture. The higher IoU indicates stronger agreement with the observer-reported second-order attention annotations. Table XIII-B further evaluates the contribution of these conditioning signals through input-removal controls.

2) Qualitative Analysis: The visualizations in Fig. 4 highlight the advantages of EASAPNet. The baseline models rely on target head orientations, limiting their ability to capture the observer's inference in emotionally or socially complex scenes. In contrast, EASAPNet integrates emotional cues, task semantics, and interaction histories to produce attention maps that are more coherent and cognitively aligned with the human reasoning process. These results are consistent with the benefit of jointly incorporating visual, emotional, and social cues for second-order attention prediction.

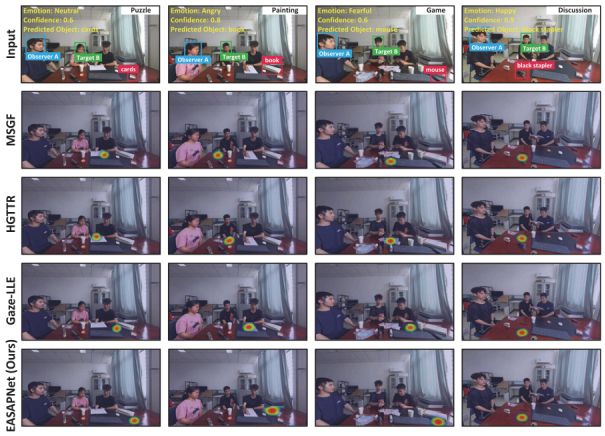


Fig. 4. Qualitative visual comparison among the attention prediction results obtained by different methods on the EASOAD.

TABLE IV

COMPONENT EVALUATION OF EASAPNET ON THE EASOAD, SHOWING THE CONTRIBUTION OF EACH PROGRESSIVELY ADDED MODULE

NO.	Components				MSE↓	KL↓	IoU↑	ECOT↑
	B	EmoEn	SIM	TAM				
1	✓				0.142	0.251	0.337	0.392
2	✓	✓			0.116	0.188	0.412	0.458
3	✓	✓	✓		0.097	0.150	0.460	0.509
4	✓	✓	✓	✓	**0.084**	**0.120**	**0.523**	**0.574**

B: Baseline **SIM:** Social Interaction Module (Fig.3-C)
EmoEn: Emotion Encoder (Fig.3-B) **TAM:** Temporal Attention Mechanism (Fig.3-D)

C. Component Evaluation

To verify the contribution of each core module contained in EASAPNet, we conduct a series of progressive component addition experiments, as shown in Table IV. The experiments start with the baseline model containing only the visual branch and gradually incorporate the emotion encoder, social interaction module, and temporal attention mechanism to evaluate the performance gain associated with each component.

The baseline model using only visual inputs (NO. 1) shows limited performance, indicating that scene semantics and object features support basic gaze prediction but cannot yield stable belief representations. Adding the emotion encoder (NO. 2) improves all the metrics, with the MSE decreasing from 0.142 to 0.116 and ECOT increasing to 0.458. These findings demonstrate that emotion modeling enhances social perception capabilities and captures attention shifts that are modulated by the observer's emotional state.

Further adding the social interaction module (NO. 3) results in continued performance improvements (MSE: 0.097; IoU: 0.460), indicating that modeling interagent relationships such as spatial proximity and shared task goals is critical for belief reasoning. Finally, incorporating the temporal attention mechanism (NO. 4) produces the best performance (MSE: 0.084; KL divergence: 0.120; IoU: 0.523; ECOT: 0.574), confirming the importance of temporal dynamics in social cognition scenarios. This mechanism allows the model to infer the current beliefs on the basis of the semantic evolution trends observed across different frames, maintaining contextual coherence and psychological consistency.

In summary, the performance improvements provided by EASAPNet follow a hierarchical pattern. Visual perception

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TABLE V

FACTOR ANALYSIS OF EASAPNET ON THE EASOAD. (A) DIFFERENT EMOTIONAL STATES; (B) DIFFERENT SOCIAL TASKS; (C) DIFFERENT MODALITIES IN TEMPORAL SOCIAL GRAPH MODELING

Emotions	MSE↓	KL↓	IoU↑	ECOT↑
Happy	**0.077**	**0.108**	**0.542**	**0.600**
Sad	0.085	0.121	0.520	0.572
Angry	0.088	0.126	0.510	0.559
Fearful	0.092	0.133	0.502	0.550
Disgusted	0.091	0.131	0.514	0.553
Surprised	0.079	0.111	0.537	0.589
Neutral	0.080	0.112	0.536	0.587

A. Different Emotions

Tasks	MSE↓	KL↓	IoU↑	ECOT↑
Puzzle	0.091	0.134	0.504	0.553
Discussion	**0.079**	**0.109**	**0.544**	**0.595**
Game	0.082	0.115	0.533	0.584
Painting	0.084	0.121	0.522	0.571
Classification	0.086	0.124	0.509	0.559
Full	0.084	0.120	0.523	0.574

B. Different Social Tasks

Types	MSE↓	KL↓	IoU↑	ECOT↑
Visual	0.132	0.208	0.410	0.425
Emotion	0.146	0.232	0.365	0.392
Task	0.139	0.220	0.380	0.401
Emotion+ Visual	0.108	0.162	0.465	0.496
Full	**0.084**	**0.120**	**0.523**	**0.574**

C. Different Modalities in Temporal Social Graph Modeling

TABLE VI

HYPERPARAMETER ANALYSIS RESULTS OBTAINED ON THE EASOAD. (A) DIFFERENT TEMPORAL STEP SIZES; (B) DIFFERENT NODE FEATURE DIMENSION; (C) DIFFERENT WEIGHTS OF EMOTION, SPATIAL, AND TASK

Step Size	MSE↓	KL↓	IoU↑	ECOT↑
5	0.097	0.146	0.465	0.498
10	**0.084**	**0.120**	**0.523**	**0.574**
15	0.091	0.130	0.489	0.522

A. Different Temporal Step Size

Feature Dimension	MSE↓	KL↓	IoU↑	ECOT↑
256	0.092	0.134	0.478	0.512
384	0.086	0.122	0.511	0.555
512	**0.084**	**0.120**	**0.523**	**0.574**

B. Different Node Feature Dimension

(E/S/T)	MSE↓	KL↓	IoU↑	ECOT↑
2/1/1	0.088	0.125	0.510	0.546
1/2/1	0.089	0.128	0.505	0.539
1/1/2	0.090	0.131	0.500	0.530
1/1/1	**0.084**	**0.120**	**0.523**	**0.574**

C. Different Weights of Emotion/Spatial/Task

provides the foundation, emotion modeling adds affective awareness, the interaction module builds interagent relations, and the temporal mechanism maintains dynamic consistency. Together, these components form a robust framework for social belief reasoning, enabling strong cognitive plausibility and emotional alignment in complex second-order attention tasks.

D. Ablation Studies

1) Impact of Emotions on Second-Order Attention Prediction: We evaluate EASAPNet under seven emotional states. As shown in Table V-A, the model exhibits clear performance variations across different affective conditions. The happy state yields the best overall results, with the lowest MSE and KL values of 0.077 and 0.108, respectively, and the highest IoU and ECOT values of 0.542 and 0.600, respectively. The surprised and neutral states also yield relatively strong performance, with ECOT scores of 0.589 and 0.587, respectively. In contrast, the negative emotional states such as anger, fear, and disgust achieve lower ECOT scores of 0.559, 0.550, and 0.553, respectively, indicating greater difficulty in terms of producing concentrated and consistent belief predictions. These results suggest that observer emotions systematically affect second-order attention inferences. By incorporating affective features, EASAPNet can adapt its belief estimation process to different emotional contexts and maintain stable performance across diverse affective conditions.

2) Effect of the Social Task Context: We examine the influence of the social task context on second-order attention prediction. As shown in Table V-B, the Discussion task provides the best overall performance, with an MSE of 0.079, a KL divergence value of 0.109, an IoU of 0.544, and an ECOT score of 0.595. The Game task also yields strong results, whereas Puzzle and Classification are more challenging, with ECOT scores of 0.553 and 0.559, respectively. This variation suggests that the task semantics and interaction structure affect the availability and reliability of social cues. Tasks with richer interpersonal interactions provide more informative gaze, orientation, and behavioral context, facilitating more consistent belief inferences. In contrast, tasks with weaker interaction cues rely more heavily on ambiguous visual evidence, increasing the level of prediction uncertainty. By incorporating task-aware representations into temporal social graph modeling, EASAPNet adapts its reasoning process to different interaction contexts and maintains robust performance across diverse social tasks.

3) Effectiveness of Temporal Social Graph Modeling: Table V-C compares unimodal and multimodal temporal social graph modeling methods. The unimodal variants show lower performance (ECOT: visual, 0.425; emotion, 0.392; task, 0.401), whereas combining emotion and visual modalities improves the results (ECOT 0.496). Incorporating all three cues, including visual, emotional, and task information, with the temporal attention mechanism in the full configuration yields the best performance (MSE: 0.084; KL divergence: 0.120; IoU: 0.523; ECOT: 0.574). These results indicate that each modality provides complementary cues: visual inputs offer direct observations, whereas emotions and tasks encode latent social priors. Their integration in the temporal graph enables dynamic beliefs to be inferred from observable behaviors, enhancing the robustness of the model to occlusions, viewpoint variations, and multiagent interactions, making multimodal temporal social graph modeling essential for obtaining coherent and cognitively plausible second-order attention predictions.

4) Sensitivity to the Temporal Step Size, Node Feature Dimensionality, and Module Weights: **1) Effect of the temporal step size.** As shown in Table VI-A, a short interval of 5 frames limits temporal dependency modeling, whereas a 10-frame interval provides the best balance between temporal consistency and responsiveness. Increasing the interval to 15 frames weakens the fine-grained behavioral cues. This confirms the importance of an appropriate temporal resolution for obtaining stable belief inferences.

2) Effect of the node feature dimensionality. Table VI-B shows that 256-dimensional features restrict the multimodal representation process, whereas 512-dimensional features capture richer emotion–vision correlations and achieve the best performance. Thus, the feature dimensionality should balance the representational capacity and computational cost of the model.

3) Balancing the module weights. As shown in Table VI-C, overemphasizing emotion, spatial, or task cues degrades the resulting performance, while the balanced setting of 1/1/1 performs best. This finding indicates that complementary multimodal cues should contribute jointly to a stable and human-aligned belief reasoning procedure.

5) Interaction Between Emotions and Spatial Distance: This experiment involves examining how emotions and spatial distance jointly affect the second-order attention prediction process. For clarity, only close- and distant-range subsets are reported. As shown in Table VII, Happy and Surprised yield the highest close-range ECOT scores of 0.617 and 0.606, respectively, whereas Angry and Fearful decrease these values to 0.493 and 0.484, respectively, at longer distances. Neutral remains relatively stable. These results suggest that the Angry, Fearful, and Disgusted conditions are more sensitive to an increasing distance.

6) User Study and Emotional Consistency Analysis: To assess the interpretability and emotional consistency of

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TABLE VII

EFFECTS OF EMOTIONS AND SPATIAL DISTANCE ON THE RESULTING PREDICTION PERFORMANCE

Distance/Emotion	Metric	Happy	Sad	Angry	Fearful	Disgusted	Surprised	Neutral
Close-range	MSE↓	0.071	0.078	0.086	0.089	0.085	0.073	0.075
	KL↓	0.101	0.115	0.127	0.132	0.124	0.108	0.106
	IoU↑	0.569	0.548	0.532	0.526	0.537	0.560	0.563
	ECOT↑	0.617	0.585	0.570	0.563	0.568	0.606	0.604
Distant-range	MSE↓	0.091	0.098	0.108	0.113	0.106	0.095	0.093
	KL↓	0.130	0.144	0.158	0.166	0.155	0.136	0.133
	IoU↑	0.494	0.476	0.448	0.437	0.451	0.487	0.490
	ECOT↑	0.553	0.526	0.493	0.484	0.497	0.542	0.546

TABLE VIII

ABSOLUTE DIFFERENCES BETWEEN THE PARTICIPANT AND MODEL PREDICTIONS UNDER DIFFERENT EMOTIONS

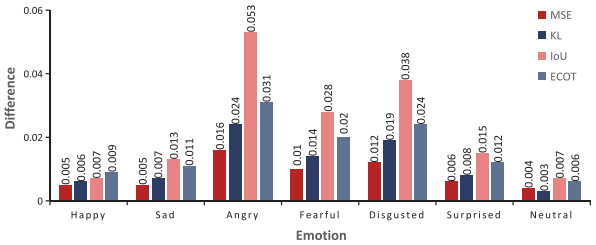


TABLE IX

FALSE-BELIEF REASONING TEST COMPARING FACTUAL AND BELIEF-BASED ATTENTION PREDICTION METHODS

Methods	Target Alignment	MSE↓	KL↓	IoU↑
Gaze-LLE	Objective Truth	0.091	0.183	0.492
	Observer Belief	0.139	0.412	0.361
EASAPNet (Ours)	Objective Truth	0.102	0.215	0.456
	Observer Belief	0.086	0.178	0.475

EASAPNet in real-world social reasoning cases, we conducted a user study in which participants watched videos of emotional interactions and inferred the gaze direction of the observed person on the basis of the observer's emotions. The model produced predictions under the same emotional conditions, and the results were compared with human gaze heatmaps. As shown in Table VIII, under stable or low-arousal states such as Happy, Sad, Surprised, and Neutral, the model–human differences were minimal (Happy: MSE, 0.005; KL divergence, 0.006; IoU, 0.007; ECOT, 0.009), indicating strong alignment between the model attention and human behavior. Under high-arousal negative emotions such as Fearful, Disgusted, and Angry, the discrepancies increased (ECOT: 0.020, 0.024, 0.031), suggesting that strong negative affects amplify attentional variations and complicate belief transitions. Despite this, EASAPNet remained stable, indicating that its emotional encoder effectively captured human-like attention modulations across different affective contexts, preserving belief coherence and emotional alignment. Overall, these results show that the model internalizes the coupling between emotions and attention, enabling a belief inference process that is consistent with the social reasoning scheme of humans.

7) Quantifying the Separation Between Subjective Beliefs and Objective Reality: To evaluate the ability of the model to distinguish subjective beliefs from objective reality, we designed a false-belief test inspired by the theory-of-mind paradigm. This experiment involved assessing whether the model could represent what was true and what the observer believed to be true. As shown in Table IX, Gaze-LLE performed well under factual alignment (MSE: 0.091, KL divergence: 0.183, IoU: 0.492) but declined under belief alignment (KL divergence: 0.412, IoU: 0.361), indicating a reliance on explicit visual cues. In contrast, EASAPNet maintained balanced results across both settings and achieved better belief-aligned performance (MSE: 0.086, KL divergence: 0.178, IoU: 0.475), demonstrating its ability to separate what is seen from what is believed. Even without visual evidence, the model produced predictions that were consistent with reasoning, bridging perceptual understanding and cognitive inference.

TABLE X

ZERO-SHOT EMOTION GENERALIZATION RESULTS PRODUCED ON THE SEVEN-EMOTION EASOAD

Methods	Emotion Type	MSE↓	KL↓	IoU↑
Gaze-LLE	Seen	0.108	0.164	0.505
	Zero-shot	0.142	0.203	0.458
EASAPNet (Ours)	Seen	0.099	0.153	0.522
	Zero-shot	0.122	0.182	0.478

8) Zero-Shot Emotion Generalization Analysis: To evaluate the affective generalization of EASAPNet, we created a zero-shot split over the seven emotion categories contained in the EASOAD. Happy, sad, disgusted, and neutral were used for training and validation purposes, whereas angry, surprised, and fearful were reserved for testing. As shown in Table X, Gaze-LLE degraded more noticeably on unseen emotions, with the IoU decreasing from 0.505 to 0.458, suggesting a stronger dependence on visual cues. EASAPNet yielded a smaller decline, with the IoU decreasing from 0.522 to 0.478 and the KL divergence measure increasing from 0.153 to 0.182, indicating better cross-emotion stability. These results suggest that EASAPNet learns shared continuous affective representations rather than memorizing emotion labels, supporting an attention inference process for unseen emotions.

9) Belief–Reality Dissociation Analysis: To examine whether second-order attention prediction extends beyond first-order gaze–target estimation, we divided test samples according to the relationships among observer A's belief label, B's gaze-based target, and B's intended target when available. The gaze-based target was determined from B's visible gaze, head pose, body orientation, and spatial relations, while the intended target was identified from task context or interaction records. Samples were considered aligned when A's belief agreed with all available References; otherwise, they were treated as dissociated cases. Dissociated samples were further categorized as belief–gaze or belief–intention dissociations, with cases containing both assigned to the latter group.

As shown in Table XI, first-order gaze baselines remained competitive on aligned samples but degraded on dissociated cases, indicating that B's gaze or head cues alone are insufficient when A's belief diverges from B's visible or intended target. EASAPNet achieved larger improvements, increasing ECOT by 0.167 over the best first-order baseline in the all-dissociated group. These results demonstrate that the proposed task captures observer-centric belief formation rather than merely predicting B's first-order gaze target.

10) Test–Retest Reliability and Intraobserver Consistency: To assess the stability of observer-reported belief labels, we recalled 30 original observers for two sessions separated by

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TABLE XI

BELIEF–REALITY DISSOCIATION ANALYSIS. GF DENOTES GAZEFOLLOW, VAT DENOTES VIDEOATTENTIONTARGET, AND EAS DENOTES EASAPNET. FOR A COMPACT PRESENTATION, WE DENOTE BELIEF–GAZE DISSOCIATIONS AS B-GAZE, BELIEF–INTENTION DISSOCIATIONS AS B-INTENT, GAZEFOLLOW AS GF, VIDEOATTENTIONTARGET AS VAT, AND EASAPNET AS EAS

Subset	Ratio	GF	VAT	EAS	Gain
Aligned	61.2%	0.536	0.552	**0.605**	+0.053
B-Gaze	23.7%	0.347	0.368	**0.536**	+0.168
B-Intent	15.1%	0.321	0.342	**0.507**	+0.165
All-Dis.	38.8%	0.337	0.358	**0.525**	+0.167

TABLE XII

TEST–RETEST RELIABILITY OF THE OBSERVER-REPORTED REFERENCE BELIEF LABELS

Task	Pairs	Agr.	κ	AC1	Conf. ICC	Conf. MAE
Puzzle	252	83.3%	0.78	0.81	0.80	0.073
Painting	252	85.3%	0.81	0.84	0.84	0.065
Game	252	82.9%	0.77	0.80	0.79	0.078
Classification	252	84.5%	0.80	0.83	0.82	0.070
Discussion	252	84.5%	0.80	0.82	0.83	0.068
Overall	1260	84.1%	0.79	0.82	0.82	0.071

7 ± 1 days. Each observer reannotated 42 social interaction clips, yielding 1,260 paired judgments balanced across seven affective conditions and five social tasks. The same videos, annotation points, target persons, candidate objects, instructions, affect categories, and viewing conditions were maintained, while trial orders were randomized and previous responses were unavailable. Affects were reinduced using category-matched FilmStim clips and verified through SAM and discrete emotion checks.

As shown in Table XII, the overall target agreement reached 84.1% (95% CI: 82.0%–86.1%), with Cohen's $\kappa = 0.79$ and Gwet's AC1 = 0.82. Confidence scores achieved ICC(A,1)= 0.82, Spearman's $\rho = 0.79$, and MAE= 0.071. Dominant-emotion agreement reached 92.0% with $\kappa = 0.89$, while SAM valence and arousal achieved ICC values of 0.86 and 0.81. High-confidence judgments showed higher agreement than medium-confidence judgments (89.0% vs. 76.5%), and mixed-effects regression confirmed this relationship (odds ratio = 2.49, 95% CI: 1.87–3.31, $p < 0.001$).

These results demonstrate temporal stability, intraobserver reproducibility, and confidence consistency. They support observer reports as operational benchmark references rather than objectively observable ground truths of internal mental states. Because each clip is tied to a specific observer-conditioned state, cross-observer agreement is not a well-defined metric-matched human ceiling; the test–retest results instead provide an intraobserver reliability reference.

11) Matched B-Centric Evidence Validation: We first compared A's belief report with B's independently submitted first-person attention report at the same time point and over the same candidate object set. The reports differed in 38.4% of the 1,378 samples. By comparison, the test–retest study conducted under the same conditions in Section V-D10 achieved 84.1% agreement, corresponding to a natural switching rate of 15.9%.

For a matched-evidence validation, the same 30 observers completed two sessions using 30 additional B videos, with 10 videos each for visual-access, emotion, and role manipulations. Within each pair, all B-centric evidence was fixed, while only one A-specific factor was changed. We also evaluated a B-centric-only model and input-removal controls without modifying EASAPNet.

TABLE XIII

MATCHED-EVIDENCE HUMAN STUDY AND MODEL CONTROLS. M/C DENOTES MANIPULATED/CONTROL PAIRS; B-AG. DENOTES AGREEMENT WITH B'S FIRST-ORDER BEHAVIORAL REFERENCE

A. Human matched-evidence study				
Factor	Switch (M/C)	JSD	B-Ag.	OR
Access	42.1/15.0%	0.216	72.7→54.3%	4.05
Emotion	29.6/15.0%	0.128	68.3→58.0%	2.34
Role	25.0/13.3%	0.104	66.0→59.3%	2.11
B. Model controls				
Model	Aligned ECOT		All-Dis. ECOT	
B-centric only	0.548		0.351	
w/o e_A	0.575		0.452	
w/o A visual/spatial	0.563		0.427	
w/o task context	0.568		0.439	
EASAPNet	0.605		0.525	

As shown in Table XIII-A, changing A's visual access, emotion, or role produced switching rates of 42.1%, 29.6%, and 25.0%, respectively, all of which were above the same-condition rates of 13.3%–15.0%. The corresponding mixed-effects odds ratios were 4.05, 2.34, and 2.11 ($p < 0.01$). Thus, A-specific conditions systematically changed the reported belief, while the B-centric evidence remained fixed.

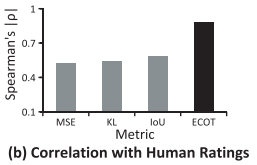
As shown in Table XIII-B, the B-centric control achieved a 0.548 ECOT score on the aligned samples but only a score of 0.351 for the dissociated samples, compared with 0.605 and 0.525 for EASAPNet, respectively. Removing observer emotions, A's visual/spatial cues or the task context reduced the dissociated ECOT scores to 0.452, 0.427, and 0.439, respectively. These results show that B-centric evidence is an important perceptual basis but does not fully determine observer-reported second-order beliefs. Role manipulation was used only for the human construct validation because the current model has no explicit role encoder.

TABLE XIV

VALIDATION OF ECOT VIA MODEL-LEVEL AND HUMAN-LEVEL CONSISTENCY ANALYSES. (A) COMPARISON ON CONVENTIONAL SPATIAL METRICS AND ECOT AMONG EASAPNET VARIANTS. (B) CORRELATION BETWEEN ECOT AND HUMAN SUBJECTIVE RATINGS

Model	MSE↓	KL↓	IoU↑	ECOT↑
Full	0.084	0.120	0.523	0.574
NoEmotion	0.089	0.130	0.517	0.425
Mismatch	0.097	0.138	0.511	0.342

(a) Comparison on conventional metrics and ECOT



(b) Correlation with Human Ratings

E. Validation of the ECOT Metric

To validate ECOT, we conducted a model probing test and a human consistency analysis. Three EASAPNet variants were constructed: (1) EASAPNet-Full with emotion encoding and belief modulation, (2) EASAPNet-NoEmotion without emotion encoding and relying on visual and social cues, and (3) EASAPNet-Mismatch with deliberately mismatched affective conditioning as a diagnostic perturbation for ECOT validation. As shown in Table XIV-(a), conventional metrics showed limited differences, whereas ECOT clearly separated the variants: the full model achieved the highest score (0.574), followed by NoEmotion (0.425) and Mismatch (0.342). This

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TABLE XV

AVERAGE PER-FRAME INFERENCE TIME (IN MILLISECONDS) OF EACH MAJOR MODULE CONTAINED IN EASAPNET

Main Steps	FPS	Milliseconds
Key Comp. 1: Visual Encoder (Sec. IV-B)		**15.94044ms**
1) *Scene feature extraction*		10.63807ms
2) *Head feature extraction*		5.20636ms
3) *Scene-head fusion*		0.09601ms
Key Comp. 2: Emotion Encoder (Sec. IV-C)		**19.72084ms**
1) *Emotion feature extraction*		19.70758ms
2) *Emotion cross-entropy loss* **(training phase only)**		0.01326ms
Key Comp. 3: Social Interaction Module (Sec. IV-D)		**1.31089ms**
1) *Node construction*		0.00451ms
2) *Edge weight computation*		0.01215ms
3) *Temporal attention mechanism*		0.09549ms
4) *Gaze prediction*		1.19874ms
Total Inference Time	**27.1**	**36.95891ms**

comparison evaluates ECOT sensitivity to affective conditioning rather than emotion-recognition accuracy.

For models without an emotion branch, the emotion was fixed to "neutral", and the distance between the true and neutral states was used as the emotional discrepancy. For non-neutral references, this assignment necessarily introduces a nonzero emotional discrepancy; hence, ECOT measures emotion-conditioned behavior rather than serving as an emotion-neutral comparator. Unlike MSE, KL divergence, and IoU, which evaluate spatial agreement, ECOT additionally measures emotional consistency and cognitive plausibility.

To examine the agreement between ECOT and human judgments, ten participants rated 28 test videos using a five-point Likert scale [57]. The mean ratings were compared with objective metrics using Spearman's rank correlation. As shown in Table XIV-(b), ECOT achieved the highest correlation with human ratings ($|\rho| = 0.88$), exceeding MSE (0.52), KL divergence (0.54), and IoU (0.58). Using the 28 video-level mean ratings, all correlations were statistically significant (MSE: $p = 0.0046$; KL: $p = 0.0030$; IoU: $p = 0.0012$; ECOT: $p < 0.001$). ECOT therefore shows the strongest observed association with human ratings, without claiming formal pairwise superiority among dependent correlations.

F. Temporal Efficiency Analysis

We evaluated the inference efficiency of EASAPNet on a workstation equipped with an Intel(R) Xeon(R) CPU and an NVIDIA GeForce RTX 4090 GPU (24 GB). As shown in Table XV, EASAPNet achieves 36.96 ms per frame (27.1 FPS), satisfying real-time requirements. The main computational cost comes from visual and emotion encoding, taking 15.94 ms and 19.72 ms, respectively, while the social interaction module requires only 1.31 ms. These results demonstrate that the lightweight social reasoning design maintains effective cognitive modeling with practical efficiency.

G. Application and Limitations Discussion

The proposed second-order attention task provides a complementary social reasoning signal rather than replacing objective gaze estimation. While first-order attention identifies what a person attends to, second-order attention models how others interpret that attention, enabling applications such as human–robot collaboration, mixed-reality teamwork, and intelligent tutoring through uncertainty-aware feedback. However, it should not be deployed as a standalone module in high-stakes scenarios and should be combined with objective gaze estimation and human-in-the-loop verification.

Several limitations remain. Observer-reported beliefs are operational references rather than objective mental-state ground truth; future work will investigate convergent validity using behavioral and physiological signals. The current task focuses on video-based interactions with specified roles, candidate objects, and confidence-filtered reports, limiting generalization to ambiguous natural scenes. The participant-disjoint protocol evaluates cross-participant but not strict cross-scene generalization. Structured scenarios, manual emotion annotations, and multimodal social reasoning introduce computational and generalization challenges. Moreover, the task does not cover the full scope of classic false-belief reasoning. Future work will explore belief–reality dissociation, weakly supervised learning, and efficient architectures.

VI. CONCLUSION

In this paper, we formulate second-order attention modeling within a visual theory-of-mind framework and define it as predicting observer-reported attentional beliefs conditioned on emotion and social context. We introduce EASOAD and propose EASAPNet, integrating visual, emotional, and social interaction representations. Experiments demonstrate that EASAPNet outperforms existing gaze-following methods and achieves stronger agreement with observer-reported beliefs. User studies and test–retest analysis further verify human alignment and label stability. This work establishes a benchmark and framework for emotion-aware belief prediction. Future work will explore self-supervised modeling and efficient multimodal integration.

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